Introduction to Financial Mathematics (Winter 2026)

Instructor: Dr. Abhishek Tilva

Course Code: SCXXX

Program / Semester: B.Tech ICT, 6^{th} Semester

Category: Science Elective

Credit Structure:

- Lecture hours per week: 3
- Tutorial hours per week: 1
- Practical hours per week: 0
- **Total Credits:** 4

Prerequisite Courses: Calculus, Probability, Linear Algebra

Course Abstract

This course introduces fundamental concepts of financial mathematics, including discrete-time models for pricing and hedging of financial derivatives, arbitrage-free valuation, and basic portfolio theory. The course then develops continuous-time models based on stochastic calculus, leading to the Black–Scholes–Merton framework.

Course Objectives

The objectives of this course are:

- To provide students with a solid foundation in the mathematical principles underlying financial engineering.
- To equip students with tools for modeling financial instruments in both discrete and continuous time.
- To develop the ability to apply stochastic methods in asset pricing.

Evaluation / Grading Policy

- Attendance: 10% (% attendance ÷ 10)
- In-semester Exam I: 25%
- In-semester Exam II: 25%
- Final Exam: 40%

Course Materials / References

Suggested Textbooks

1. Capiński, Marek, and Tomasz Zastawniak. *Mathematics for Finance: An Introduction to Financial Engineering*. Springer, 2003.

2. Petters, Arlie O., and Xiaoying Dong. *An Introduction to Mathematical Finance with Applications*. Springer, 2016.

3. Elliott, Robert J., and P. Ekkehard Kopp. *Mathematics of Financial Markets*. Springer, 2005.

4. Buchanan, J. Robert. *An Undergraduate Introduction to Financial Mathematics*. 2012.

Detailed Course Contents*

*The contents are subject to change based on time and other factors.

Topic	Content	Lectures
Foundations of Financial Markets	Basic concepts of financial instruments and markets; no-arbitrage principle; one-step binomial model; risk and return; forward contracts; introduction to options.	3
Riskless Assets	Time value of money; simple and compound interest; annuities and perpetuities; continuous compounding; zero-coupon and coupon bonds; money market account.	4
Risky Assets	Dynamics of stock prices; binomial tree model; risk-neutral probability and martingale property; trinomial tree model and continuous-time limit.	3
Discrete-Time Market Models	Stock and money market models; investment strategies; principle of no-arbitrage; fundamental theorem of asset pricing.	5
Portfolio Management	Risk of securities; portfolios with two or several securities; expected return and variance of portfolios; Capital Asset Pricing Model (CAPM); capital market line; beta factor and security market line.	5
Derivative Contracts	Forward and futures contracts: pricing and hedging applications; options: European and American options; put-call parity; properties of calls; time value of options.	5
Option Pricing	European and American options in the binomial model; one-step and multi-step models; Cox–Ross–Rubinstein formula; Black–Scholes formula.	5
Financial Engineering	Delta-hedging and Greeks; Value-at-Risk and business risk; case studies on hedging and speculation with derivatives.	3
Stochastic Calculus	Brownian motion, filtrations, stopping times; stochastic integral; Itô's lemma; applications to modeling stock prices.	6
Continuous-Time Finance	Black–Scholes–Merton model; risk-neutral valuation; comparison with discrete-time models.	6